

Mudit Mehta

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Experience

Addverb Technologies

Robotics Intern

Noida, Uttar Pradesh (On-site)

Jun 2025 - Aug 2025

- Built and deployed a customer-facing error monitoring dashboard for the live Fleet Management System, including REST APIs, backend services, and third-party integrations used by robot operators in production.
- Reduced localization drift in autonomous mobile robots by implementing IMU and odometry sensor fusion using an Extended Kalman Filter (EKF) within a ROS-based navigation stack.
- Debugged and resolved issues across the FMS codebase in an Agile team environment, participating in daily scrums.
- Keywords:** C++, TypeScript, Docker, MongoDB, Redis, AngularJS

Wipro Limited

Project Intern

Bengaluru, Karnataka (Hybrid)

Jun 2024 – Aug 2024

- Project: Software Installation Agent** (Bound by NDA)
- Built an internal AI agent using LangChain and OpenAI that recommends relevant proprietary applications based on natural language queries, enabling on-the-go access to matched tools.
- Integrated an Azure Vector Database as the retrieval backend, enabling semantic search across a catalogue of applications.
- Keywords:** LangChain, OpenAI, Azure Vector Database, Python

HP, Inc.

Intern

Gurugram, Haryana (On-site)

Jul 2023 – Aug 2023

- Project: AI in Retail**
- Researched and evaluated AI tool providers for customer tracking and store analytics across 700+ HP World retail locations, producing a shortlist of implementation-ready vendors.

Swar Kala Sangam

Software Development Intern

Gurugram, Haryana (On-site)

Jan 2022 – Jul 2022

- Deployed and maintained a production web application on AWS, resolving 15+ backend support issues to sustain uptime.
- Keywords:** HTML, CSS, JavaScript, AWS

Projects

ShopAssist

Sep 2025 – Mar 2026

- Project Repository:** <https://github.com/muditmehta07/ShopAssist>
- ShopAssist is an Autonomous Shopper Assistance System which aims to provide an effortless shopping experience to customers in retail environments. Disguised as a shopping cart, it navigates autonomously to lead the user to their desired products or follow them around, removing the physical hassle of handling a cart.
- Achieved >95% navigation success rate across dynamic, high-traffic, and low-light scenarios by optimizing ROS 2 Nav2 and SLAM Toolbox parameters within Gazebo-simulated environments.
- Reduced system latency to <20ms for real-time 2D occupancy mapping and live video telemetry through a high-performance ReactJS dashboard and optimized communication protocols.
- Keywords:** ROS2, Gazebo, Rviz2, Nav2, LangChain, Ollama, FastAPI, MongoDB, ReactJS, Docker, Podman, Python

Publications

Robust Localization and Mapping of Autonomous Robots in Challenging Spaces via Thermal-LiDAR Fusion

Awaiting Publish

- Associated with ShopAssist.
- Co-authored with Dr. Sunil Pathak (Head of Department, Amity School of Engineering & Technology, Rajasthan)
- Accepted at ICNWC'26 – SRM Institute of Science and Technology.

Education

Amity School of Engineering & Technology, Amity University

Bachelor of Technology Computer Science & Engineering

Jaipur, Rajasthan

Sep 2022 - Jun 2026

- Minor Degree in Bioinformatics

Certifications

Salesforce

Salesforce Certified AI Associate

Credential ID: 5704656

Issued: Jan 2025

Skills

AI & Robotics

LangChain, TensorFlow, Flask, FastAPI, Ollama, NumPy, Pandas, ROS2, Rviz2, Nav2, SLAM, Gazebo

Languages & Tools

Python, C/C++, TypeScript, Docker, Podman, MongoDB, Redis, Git/GitHub